CUSTOMER CHURN PREDICTION & LIFETIME VALUE (LTV) ENGINE

PROJECT REPORT

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ABSTRACT

Customer retention has become one of the most important goals for businesses, especially in subscription-based industries such as telecommunications. Understanding why customers leave and identifying customers who are likely to churn enables organizations to develop better retention strategies and improve long-term profitability.

The **Customer Churn Prediction & Lifetime Value (LTV) Engine** is an end-to-end data analytics and machine learning project developed to analyze customer behavior, predict customer churn, estimate customer lifetime value, and present actionable business insights through an interactive Power BI dashboard.

The project began with collecting and preparing customer data by performing data cleaning, handling missing values, removing duplicates, and conducting exploratory data analysis. Feature engineering techniques were then applied to improve the quality of the dataset before training multiple machine learning models.

Three machine learning algorithms—Logistic Regression, Decision Tree, and Random Forest—were developed and evaluated using various performance metrics such as Accuracy, ROC-AUC Score, Confusion Matrix, and Classification Report. Customer Lifetime Value (LTV) was also calculated to estimate the future value of each customer. Based on the calculated LTV, customers were segmented into Bronze, Silver, Gold, and Platinum categories to support business decision-making.

To make the prediction model accessible for real-world applications, the trained model was deployed using FastAPI. REST API endpoints were developed to support both single-customer and batch predictions, allowing users to interact with the machine learning model efficiently.

Finally, an interactive four-page Power BI dashboard was created to visualize customer demographics, churn behavior, service usage, customer lifetime value, revenue analysis, and key performance indicators. The dashboard transforms complex analytical results into meaningful business insights that can help organizations identify high-risk customers and improve customer retention strategies.

Overall, this project demonstrates the complete lifecycle of a machine learning solution, from data preprocessing and predictive modeling to deployment and business intelligence reporting.

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CHAPTER 1

INTRODUCTION

In today's highly competitive business environment, customer retention has become just as important as acquiring new customers. Organizations invest significant resources in attracting customers, but retaining existing customers often leads to greater profitability and long-term business growth. Understanding customer behavior and identifying the factors that contribute to customer churn allows businesses to develop targeted strategies that improve customer satisfaction and reduce revenue loss.

Customer churn refers to the situation where customers discontinue using a company's products or services. High churn rates can significantly affect a company's revenue, customer base, and overall market competitiveness. Therefore, predicting customer churn before it happens has become an essential application of machine learning and data analytics.

This project, titled **"Customer Churn Prediction & Lifetime Value (LTV) Engine,"** was developed to build a complete business intelligence solution that combines data analytics, machine learning, API deployment, and interactive dashboard visualization. Rather than focusing only on prediction, the project provides a complete analytical workflow that supports informed business decision-making.

The project begins with collecting and preparing customer data through preprocessing techniques such as handling missing values, removing duplicate records, correcting data types, and performing exploratory data analysis. These preprocessing steps improve data quality and provide a strong foundation for predictive modeling.

After preprocessing, multiple machine learning algorithms are trained to predict whether a customer is likely to churn. Model performance is evaluated using standard classification metrics to identify the most suitable algorithm for prediction.

In addition to churn prediction, the project estimates **Customer Lifetime Value (LTV)**, which measures the expected long-term value generated by each customer. Based on the estimated LTV, customers are segmented into different value categories, allowing businesses to prioritize customer retention strategies effectively.

To make the predictive model practical for business use, it is deployed using FastAPI, enabling users to access predictions through REST API endpoints. Finally, all analytical findings are presented using an interactive Power BI dashboard that allows business users to explore customer behavior through dynamic visualizations and key performance indicators.

This project demonstrates the integration of data science, machine learning, software deployment, and business intelligence into a single end-to-end solution capable of supporting real-world customer retention strategies.

CHAPTER 2

PROBLEM STATEMENT

Customer churn is one of the major challenges faced by subscription-based businesses. Losing customers not only reduces revenue but also increases the cost of acquiring new customers. Many organizations possess large amounts of customer data; however, without proper analysis, it becomes difficult to identify customers who are likely to leave and understand the reasons behind their decisions.

Traditional methods of customer analysis often rely on historical reports and manual interpretation, making it challenging to identify complex relationships within customer data. Businesses require intelligent systems that can automatically analyze customer information, predict future churn, estimate customer value, and provide meaningful insights for strategic planning.

The objective of this project is to develop a comprehensive analytical solution capable of predicting customer churn, estimating Customer Lifetime Value (LTV), segmenting customers based on their value, deploying prediction services through REST APIs, and presenting business insights using an interactive Power BI dashboard. Such a solution enables organizations to proactively retain valuable customers and improve overall business performance.

CHAPTER 3

PROJECT OBJECTIVES

The primary objective of this project is to build an end-to-end Customer Churn Prediction and Lifetime Value Engine that assists businesses in understanding customer behavior and improving customer retention strategies.

The specific objectives of this project are:

- To collect and preprocess customer data for analysis.
- To clean and transform raw data into a machine-learning-ready dataset.
- To perform Exploratory Data Analysis (EDA) and identify important customer behavior patterns.
- To engineer meaningful features that improve prediction accuracy.
- To develop machine learning models capable of predicting customer churn.
- To evaluate different machine learning algorithms using standard performance metrics.
- To estimate Customer Lifetime Value (LTV) for every customer.
- To segment customers into Bronze, Silver, Gold, and Platinum categories based on their lifetime value.
- To deploy the trained prediction model using FastAPI.
- To create REST API endpoints for single and batch customer predictions.
- To design an interactive Power BI dashboard that visualizes churn trends, customer behavior, customer lifetime value, and business performance.

- To generate meaningful business insights that support customer retention and strategic decision-making.

CHAPTER 4

SCOPE OF THE PROJECT

This project covers the complete lifecycle of a customer churn prediction system, beginning with raw data preparation and ending with business intelligence reporting.

The project includes data preprocessing, exploratory data analysis, feature engineering, predictive modeling, customer lifetime value estimation, customer segmentation, REST API deployment using FastAPI, and dashboard development using Microsoft Power BI.

The developed solution can assist businesses in identifying customers who are at high risk of leaving, understanding customer behavior, estimating long-term customer value, and supporting data-driven decision-making through interactive dashboards.

Although the project is developed using a telecom customer dataset, the same methodology can be applied to other industries such as banking, insurance, e-commerce, healthcare, and subscription-based services.

CHAPTER 5

DATASET DESCRIPTION

The dataset used in this project contains customer information collected from a telecommunications company. It includes demographic information, subscribed services, contract details, billing information, and customer churn status. The dataset serves as the foundation for both predictive modeling and business intelligence analysis.

The dataset contains a combination of categorical and numerical attributes, allowing various analytical techniques to be applied during different stages of the project.

The major attributes included in the dataset are:

- Customer ID
- Gender

- Senior Citizen
- Partner
- Dependents
- Tenure
- Phone Service
- Multiple Lines
- Internet Service
- Online Security
- Online Backup
- Device Protection
- Tech Support
- Streaming TV
- Streaming Movies
- Contract Type
- Paperless Billing
- Payment Method
- Monthly Charges
- Total Charges
- Customer Churn Status

These variables provide valuable information for analyzing customer behavior, building predictive models, estimating customer lifetime value, and generating business insights through visualization.

CHAPTER 6

WEEK 1 – DATA COLLECTION, DATA PREPROCESSING AND EXPLORATORY DATA ANALYSIS

6.1 Introduction

The first phase of this project focused on understanding the dataset and preparing it for further analysis. Before building any machine learning model, it is important to ensure that the data is accurate, complete, and suitable for analysis. Poor-quality data can negatively impact the performance of predictive models and lead to incorrect business insights.

During this stage, the dataset was explored to understand its structure, identify missing values, detect duplicate records, examine data types, and analyze the relationships between different customer attributes. Various preprocessing techniques were applied to improve the quality of the data before moving to machine learning.

In addition to data cleaning, Exploratory Data Analysis (EDA) was performed to identify patterns and trends within the customer data. These insights helped in understanding customer behavior and identifying potential factors contributing to customer churn.

6.2 Data Collection

The project uses a telecom customer dataset containing information about customer demographics, subscribed services, billing details, contract information, and churn status.

The dataset includes both numerical and categorical features, making it suitable for machine learning classification and business intelligence analysis.

The collected data includes information such as:

- Customer demographic details
- Service subscriptions
- Contract information
- Payment methods
- Monthly charges
- Total charges
- Customer tenure
- Churn status

These attributes provide a comprehensive view of customer behavior and serve as the basis for predictive analytics throughout the project.

6.3 Understanding the Dataset

Before performing any preprocessing, the dataset was carefully examined to understand its overall structure and contents.

The following activities were carried out:

- Loaded the dataset into Python.
- Displayed the first few records to understand the data structure.
- Examined the number of rows and columns.
- Identified numerical and categorical variables.
- Checked data types for each feature.

- Generated summary statistics.
- Reviewed column names and feature descriptions.

This initial analysis helped identify the preprocessing steps required before building predictive models.

6.4 Data Cleaning

Data cleaning is one of the most important stages in any data analytics project. Raw datasets often contain inconsistencies, missing values, duplicate records, or incorrect data types that can affect the overall analysis.

Several preprocessing techniques were applied to improve the quality of the dataset.

Handling Missing Values

The dataset was examined for missing and blank values. Missing information was identified and handled appropriately to ensure that the dataset remained complete and suitable for analysis.

Blank entries, especially in numerical columns, were converted into valid numerical values wherever necessary. This helped prevent errors during feature engineering and model training.

Removing Duplicate Records

Duplicate customer records can introduce bias into machine learning models by giving additional weight to repeated observations.

The dataset was checked for duplicate entries, and duplicate records were removed to ensure that each customer was represented only once.

Correcting Data Types

Several columns required data type conversion before analysis.

Numerical columns such as tenure, monthly charges, and total charges were verified to ensure they were stored as numerical values.

Categorical columns such as gender, contract type, internet service, and payment method were stored using appropriate categorical data types for easier analysis.

Data Consistency

The dataset was reviewed to ensure consistency across all variables.

This process included:

- Checking unique values in categorical columns.
- Verifying category labels.
- Identifying inconsistent spellings or formatting.
- Ensuring logical consistency between related variables.

These steps improved the reliability of the dataset and reduced the possibility of inaccurate results during later stages of the project.

6.5 Exploratory Data Analysis (EDA)

After completing data cleaning, Exploratory Data Analysis (EDA) was performed to understand customer behavior and identify meaningful patterns within the dataset.

EDA is an important step because it helps reveal relationships between variables before predictive models are developed.

Several visualizations and statistical summaries were created to better understand customer characteristics and churn behavior.

The analysis focused on identifying trends related to demographics, services, billing information, and customer retention.

6.6 Customer Demographic Analysis

Customer demographics were analyzed to understand how different customer groups contribute to churn.

The following demographic variables were studied:

- Gender
- Senior Citizen
- Partner Status
- Dependents

The analysis showed that customer churn varies across different demographic groups, although some demographic variables had only a moderate impact compared to service-related features.

Understanding customer demographics provides businesses with valuable information for developing targeted customer retention strategies.

6.7 Contract Type Analysis

Customer contract type plays a significant role in determining customer retention.

The dataset was analyzed to compare churn rates across different contract categories.

The analysis indicated that customers with month-to-month contracts tend to leave the company more frequently than customers with long-term contracts.

Longer contract durations generally contribute to improved customer retention because customers remain committed for extended periods.

6.8 Service Usage Analysis

The dataset contains information about several additional services offered by the telecom company.

These include:

- Online Security
- Online Backup
- Device Protection
- Tech Support
- Streaming TV
- Streaming Movies
- Phone Service
- Multiple Lines

Each service was analyzed individually to understand its relationship with customer churn.

Customers who subscribed to value-added services such as Online Security and Tech Support generally demonstrated lower churn rates than customers who did not use these services.

These findings suggest that additional service offerings may improve customer loyalty.

6.9 Billing and Payment Analysis

Customer billing information was also examined to identify financial factors influencing churn.

The analysis focused on:

- Monthly Charges
- Total Charges
- Payment Methods
- Paperless Billing

The results indicated that customers with relatively higher monthly charges were more likely to discontinue their services.

Similarly, certain payment methods showed higher churn rates than others, suggesting that billing preferences may influence customer retention.

6.10 Outlier Detection

Outlier detection was performed to identify unusually high or low values within numerical variables.

Box plots and statistical methods were used to examine variables such as:

- Monthly Charges
- Total Charges
- Tenure

Detecting outliers helps improve data quality and provides a better understanding of unusual customer behavior.

Rather than removing all outliers, they were carefully examined to determine whether they represented valid business scenarios.

6.11 Statistical Analysis

Basic statistical analysis was conducted to summarize the numerical variables in the dataset.

The following statistical measures were calculated:

- Mean
- Median
- Standard Deviation
- Minimum Value
- Maximum Value
- Quartiles

These measures helped describe the overall distribution of customer data and identify variations among different customer groups.

6.12 Business Insights Obtained from EDA

The exploratory analysis generated several important business insights that guided the remaining stages of the project.

Some of the major observations include:

- Customers with month-to-month contracts have a higher probability of churn.
- Higher monthly charges are associated with increased customer churn.
- Customers with longer tenure generally remain loyal to the company.
- Customers using Online Security and Tech Support services are less likely to churn.
- Customers using Electronic Check as a payment method exhibit relatively higher churn.
- Fiber Optic Internet customers tend to have higher churn compared to DSL users.
- Additional value-added services contribute positively to customer retention.

These findings provided valuable guidance for feature engineering and machine learning model development in the next phase of the project.

6.13 Outcome of Week 1

The first phase of the project successfully established a strong foundation for predictive modeling. The dataset was cleaned, validated, and transformed into a high-quality dataset suitable for machine learning.

Comprehensive exploratory data analysis helped uncover meaningful customer behavior patterns and identify the factors most closely associated with customer churn. These insights were later used to guide feature engineering and model development.

By the end of Week 1, the project achieved the following milestones:

- Successfully loaded and explored the customer dataset.

- Cleaned and preprocessed the data.
- Handled missing values and duplicate records.
- Verified data quality and consistency.
- Performed exploratory data analysis.
- Identified important customer churn patterns.
- Generated business insights for predictive modeling.

The completion of these tasks ensured that the dataset was ready for feature engineering, machine learning model development, and Customer Lifetime Value (LTV) estimation in the following phase of the project.

CHAPTER 7

WEEK 2 – FEATURE ENGINEERING, MACHINE LEARNING, CUSTOMER LIFETIME VALUE (LTV) AND CUSTOMER SEGMENTATION

7.1 Introduction

After completing data cleaning and exploratory data analysis, the next phase of the project focused on preparing the dataset for predictive modeling. Machine learning algorithms require numerical and properly scaled data to achieve better performance. Therefore, several feature engineering techniques were applied before training the models.

The objective of this phase was to develop accurate machine learning models capable of predicting customer churn while also estimating Customer Lifetime Value (LTV). In addition, customers were segmented into different value categories to help businesses prioritize customer retention efforts.

This phase combined data preprocessing, predictive analytics, and customer segmentation to build a comprehensive decision-support system.

7.2 Feature Engineering

Feature engineering is the process of transforming raw data into meaningful features that improve the performance of machine learning models. Proper feature engineering helps algorithms better understand relationships within the data and improves prediction accuracy.

Several preprocessing techniques were applied before model training.

The feature engineering process included:

- Encoding categorical variables
- Scaling numerical features
- Selecting relevant predictor variables
- Preparing input and target datasets
- Splitting data into training and testing sets

These steps ensured that the dataset was suitable for machine learning algorithms.

7.3 Encoding Categorical Variables

The telecom dataset contains several categorical variables such as gender, contract type, internet service, payment method, and customer services.

Since machine learning algorithms cannot directly process categorical values, these variables were converted into numerical representations using encoding techniques.

Encoding allowed the models to interpret customer attributes mathematically while preserving the relationships between different categories.

This transformation ensured compatibility with all machine learning algorithms used in the project.

7.4 Feature Scaling

The dataset contains numerical variables with different ranges, including tenure, monthly charges, and total charges.

Machine learning models may perform poorly when numerical features have significantly different scales. Therefore, feature scaling was applied to normalize these values and improve model performance.

Scaling helps prevent variables with larger numerical ranges from dominating the learning process and contributes to faster model convergence.

7.5 Training and Testing Dataset

After preprocessing, the dataset was divided into two subsets:

- Training Dataset
- Testing Dataset

The training dataset was used to train the machine learning models, while the testing dataset was reserved for evaluating model performance on unseen data.

Separating the dataset into training and testing sets helps measure how well the models generalize to new customer records and reduces the risk of overfitting.

CHAPTER 8

MACHINE LEARNING MODELS

8.1 Introduction

The primary objective of the machine learning phase was to develop predictive models capable of identifying customers who are likely to churn.

Multiple classification algorithms were developed and compared to determine which model produced the most reliable predictions.

Using multiple algorithms allows a comprehensive comparison of prediction performance and helps identify the most suitable model for deployment.

The following machine learning algorithms were implemented:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier

Each model was trained using the prepared dataset and evaluated using standard classification metrics.

8.2 Logistic Regression

Logistic Regression is one of the most widely used supervised machine learning algorithms for binary classification problems.

In this project, Logistic Regression was used to classify customers into two categories:

- Customer Will Churn
- Customer Will Not Churn

The algorithm estimates the probability of customer churn based on customer characteristics such as contract type, monthly charges, tenure, payment method, and subscribed services.

Logistic Regression provides a strong baseline model because it is simple, interpretable, and computationally efficient.

8.3 Decision Tree Classifier

The Decision Tree algorithm classifies customers by creating a hierarchical structure of decision rules based on customer attributes.

Each internal node represents a decision, while each branch represents an outcome of that decision.

The model identifies important customer characteristics that contribute to churn and creates decision paths accordingly.

One of the main advantages of Decision Trees is their interpretability, making it easier for businesses to understand the reasoning behind predictions.

8.4 Random Forest Classifier

Random Forest is an ensemble machine learning algorithm that combines multiple Decision Trees to improve prediction accuracy.

Instead of relying on a single tree, Random Forest builds numerous trees using different subsets of the dataset and combines their predictions.

This approach reduces overfitting and generally produces more accurate and robust predictions compared to a single Decision Tree.

Among the implemented algorithms, Random Forest demonstrated strong predictive capabilities and was selected as one of the most reliable models for customer churn prediction.

CHAPTER 9

MODEL EVALUATION

9.1 Introduction

After training the machine learning models, their performance was evaluated using standard classification metrics.

Model evaluation helps determine how accurately each algorithm predicts customer churn and assists in selecting the most suitable model for deployment.

Several evaluation techniques were applied during this stage.

9.2 Accuracy

Accuracy measures the proportion of correctly classified customer records.

It represents the percentage of customers whose churn status was predicted correctly by the model.

Higher accuracy indicates better predictive performance; however, accuracy alone is not always sufficient for evaluating classification models, especially when datasets contain class imbalance.

9.3 ROC-AUC Score

The Receiver Operating Characteristic – Area Under Curve (ROC-AUC) score measures the ability of a classification model to distinguish between churned and retained customers.

A higher ROC-AUC score indicates better discrimination between the two classes and generally reflects stronger predictive performance.

This metric provides a more balanced evaluation than accuracy when dealing with classification problems.

9.4 Confusion Matrix

A Confusion Matrix provides a detailed summary of prediction results.

It compares predicted outcomes with actual customer churn status and consists of four components:

- True Positives
- True Negatives
- False Positives
- False Negatives

The confusion matrix helps identify the types of prediction errors made by each machine learning model.

9.5 Classification Report

The Classification Report summarizes several important evaluation metrics, including:

- Precision
- Recall
- F1-Score
- Support

These metrics provide deeper insight into model performance and help evaluate prediction quality beyond overall accuracy.

9.6 Model Comparison

The three machine learning algorithms were compared using multiple evaluation metrics.

The comparison considered:

- Prediction Accuracy
- ROC-AUC Score
- Classification Report
- Confusion Matrix

Based on the evaluation results, the best-performing model was selected for deployment in the subsequent phase of the project.

Evaluating multiple algorithms ensured that the deployed model offered reliable customer churn predictions.

CHAPTER 10

CUSTOMER LIFETIME VALUE (LTV)

10.1 Introduction

Customer Lifetime Value (LTV) represents the estimated revenue that a customer is expected to generate throughout their relationship with the company.

Predicting churn alone is not sufficient for business decision-making. Businesses must also identify which customers contribute the highest long-term value.

Calculating Customer Lifetime Value allows organizations to prioritize valuable customers and allocate retention resources more effectively.

10.2 LTV Calculation

Customer Lifetime Value was estimated using customer billing information and tenure-related variables.

The calculation considered factors such as:

- Monthly Charges
- Customer Tenure
- Customer Spending Patterns

The resulting LTV values provide an estimate of each customer's long-term contribution to business revenue.

10.3 Importance of Customer Lifetime Value

Estimating Customer Lifetime Value provides several business advantages.

It helps organizations:

- Identify high-value customers.
- Improve customer retention strategies.
- Allocate marketing budgets efficiently.
- Increase long-term profitability.
- Support customer relationship management.

By combining churn prediction with Customer Lifetime Value, businesses can focus their efforts on retaining customers who contribute the greatest long-term value.

CHAPTER 11

CUSTOMER SEGMENTATION

11.1 Introduction

Customer segmentation divides customers into groups based on shared characteristics.

In this project, customers were segmented according to their estimated Lifetime Value.

This approach allows businesses to develop personalized retention strategies for different customer groups.

11.2 Customer Categories

Customers were classified into four categories:

Bronze Customers

Bronze customers represent the lowest lifetime value segment.

These customers generally contribute lower overall revenue and require cost-effective retention strategies.

Silver Customers

Silver customers contribute moderate business value.

Although they are not among the highest-value customers, they still represent an important customer segment with growth potential.

Gold Customers

Gold customers generate high business value and demonstrate strong long-term relationships with the company.

Businesses should focus on maintaining customer satisfaction and providing personalized services for this segment.

Platinum Customers

Platinum customers represent the most valuable customer segment.

They generate the highest revenue and significantly contribute to long-term business profitability.

Retaining Platinum customers should be a top priority through premium customer support, loyalty programs, and personalized offers.

11.3 Business Benefits of Customer Segmentation

Customer segmentation enables businesses to:

- Prioritize customer retention.
- Develop personalized marketing campaigns.
- Improve customer satisfaction.
- Allocate resources more efficiently.
- Increase overall profitability.
- Strengthen long-term customer relationships.

Combining Customer Lifetime Value with customer segmentation provides organizations with valuable strategic insights that support informed business decisions.

11.4 Outcome of Week 2

Week 2 successfully transformed the cleaned dataset into a complete predictive analytics solution.

During this phase, feature engineering techniques were applied to prepare the dataset for machine learning. Multiple classification algorithms were trained and evaluated to predict customer churn accurately. In addition, Customer Lifetime Value (LTV) was estimated, and customers were segmented into Bronze, Silver, Gold, and Platinum categories based on their value to the business.

By the end of Week 2, the following objectives were achieved:

- Encoded categorical variables.
- Applied feature scaling.
- Performed feature engineering.
- Built Logistic Regression, Decision Tree, and Random Forest models.
- Evaluated models using Accuracy, ROC-AUC Score, Confusion Matrix, and Classification Report.
- Calculated Customer Lifetime Value (LTV).
- Segmented customers into Bronze, Silver, Gold, and Platinum categories.
- Prepared the best-performing model for deployment.

This phase established the predictive intelligence component of the project and laid the foundation for deploying the machine learning model as a REST API in the following week.

CHAPTER 12

WEEK 3 – MODEL DEPLOYMENT USING FASTAPI

12.1 Introduction

After successfully building and evaluating the machine learning models, the next step was to make the prediction model accessible for real-world use. A machine learning model becomes much more valuable when it can be integrated into applications that allow users to make predictions without needing to run Python scripts manually.

To achieve this, the trained churn prediction model was deployed using **FastAPI**, a modern Python framework for building high-performance REST APIs. FastAPI enables developers to expose machine learning models through HTTP endpoints, making it possible for external applications to send customer information and receive churn predictions instantly.

During this phase, the trained model was saved using Pickle, REST API endpoints were developed, input validation was implemented, and the API was tested using Swagger UI.

12.2 Why FastAPI?

FastAPI is a modern web framework designed for building APIs quickly and efficiently. It provides excellent performance, automatic API documentation, and built-in data validation.

The main reasons for selecting FastAPI in this project include:

- High performance and speed.
- Easy integration with machine learning models.
- Automatic API documentation.
- Built-in request validation.
- Easy testing through Swagger UI.
- Simple deployment and scalability.

These features make FastAPI an ideal choice for deploying machine learning models into production environments.

12.3 Saving the Machine Learning Model

Once the best-performing machine learning model was identified during Week 2, it was saved using the **Pickle** library.

Saving the trained model eliminates the need to retrain it every time predictions are required. Instead, the saved model can simply be loaded into memory and used for making predictions.

The serialized model preserves all learned parameters and enables efficient deployment through the FastAPI application.

12.4 REST API Development

The churn prediction model was deployed as a REST API using FastAPI.

The API acts as an interface between users and the machine learning model. Users send customer information to the API, and the API processes the data using the trained model before returning the prediction.

The API architecture follows a simple workflow:

User Request → FastAPI Application → Trained Machine Learning Model → Prediction Response

This architecture allows external applications to interact with the prediction model without requiring knowledge of the underlying machine learning implementation.

12.5 API Endpoints

Several REST API endpoints were developed during this phase to provide different functionalities.

The API supports:

- Health check endpoint.
- Single customer prediction.
- Batch customer prediction.

Each endpoint performs a specific task and returns structured responses in JSON format.

This modular approach improves maintainability and allows future enhancements without affecting existing functionality.

12.6 Single Customer Prediction

One of the primary features of the API is the ability to predict churn for an individual customer.

Users provide customer information such as demographic details, subscribed services, contract information, and billing details through an API request.

The FastAPI application validates the input data, passes it to the trained machine learning model, and returns the predicted churn status.

This feature enables businesses to evaluate the churn risk of individual customers in real time.

12.7 Batch Customer Prediction

In addition to individual predictions, the API also supports batch prediction.

Instead of processing one customer at a time, users can submit multiple customer records within a single request.

The API processes each record sequentially and returns predictions for all customers.

Batch prediction is particularly useful for organizations that need to analyze large customer datasets efficiently.

12.8 Input Validation

One of the key advantages of FastAPI is its automatic input validation.

Before generating predictions, the API verifies that all required customer information is provided and that the values match the expected data types.

Input validation helps prevent invalid requests from reaching the machine learning model and ensures reliable prediction results.

Examples of validation include:

- Checking numerical fields.
- Verifying required customer attributes.
- Rejecting incomplete requests.
- Returning meaningful error messages for invalid inputs.

This improves the robustness and reliability of the deployed application.

12.9 JSON Request and Response

Communication between the client application and the FastAPI server takes place using JSON.

A request contains customer information in JSON format, while the API returns the prediction result as a JSON response.

Using JSON provides several benefits:

- Platform-independent communication.

- Easy integration with web and mobile applications.
- Human-readable data format.
- Lightweight message exchange.

JSON has become the standard format for REST APIs and supports seamless integration across different software platforms.

12.10 Swagger UI Testing

FastAPI automatically generates interactive API documentation using Swagger UI.

Swagger UI provides a web-based interface where users can:

- View available API endpoints.
- Understand request and response formats.
- Test API functionality.
- Submit sample requests.
- View prediction results instantly.

Using Swagger UI significantly simplified API testing during development and ensured that all endpoints functioned correctly.

12.11 Project Organization

During deployment, the project files were organized into a structured directory to improve readability and maintainability.

The project included separate components for:

- Machine learning model
- API application
- Dataset
- Supporting files
- Documentation

A well-organized project structure makes future maintenance, updates, and collaboration easier.

12.12 Version Control Using Git and GitHub

Throughout the project, Git and GitHub were used for version control and project management.

Changes made during development were tracked using Git commits, allowing the project to maintain a clear development history.

GitHub served as a centralized repository where the complete project was stored, including source code, documentation, machine learning components, FastAPI application, and Power BI dashboard.

Using version control also made it easier to manage updates and maintain different stages of project development.

12.13 Advantages of API Deployment

Deploying the churn prediction model as a REST API provides several practical advantages.

Some of the major benefits include:

- Real-time customer churn prediction.
- Easy integration with external software applications.
- Faster prediction process.
- Improved scalability.
- Platform-independent accessibility.
- Centralized model management.
- Simplified deployment and maintenance.

These advantages make the solution suitable for real-world business environments where predictive analytics must be accessible to multiple users or applications.

12.14 Business Applications

The deployed API can be integrated into various business systems.

Some practical applications include:

- Customer Relationship Management (CRM) systems.
- Telecom customer support platforms.
- Customer retention applications.
- Business intelligence dashboards.
- Internal analytics platforms.
- Mobile and web applications.

Through API integration, businesses can automatically predict customer churn and initiate retention strategies before customers decide to leave.

12.15 Outcome of Week 3

Week 3 successfully transformed the machine learning model into a deployable business solution.

The trained churn prediction model was saved using Pickle and deployed using FastAPI. REST API endpoints were created for both single and batch customer predictions. Input validation was implemented to improve reliability, and Swagger UI was used to test the API and verify its functionality. The project files were organized systematically, and GitHub was used to maintain version control and documentation.

By the end of Week 3, the following milestones had been achieved:

- Saved the trained machine learning model using Pickle.
- Developed a FastAPI application.
- Created REST API endpoints.
- Implemented single customer prediction.
- Implemented batch customer prediction.
- Added request validation.
- Tested the API using Swagger UI.
- Organized the project structure.
- Uploaded the completed work to GitHub.

Completing this phase converted the predictive model into a practical application that can be integrated into real-world business systems, paving the way for the final stage of the project, where business insights were presented through an interactive Power BI dashboard.

CHAPTER 13

WEEK 4 – POWER BI DASHBOARD DEVELOPMENT

13.1 Introduction

After successfully completing data preprocessing, machine learning model development, Customer Lifetime Value (LTV) estimation, customer segmentation, and API deployment, the final stage of the project focused on presenting the results through an interactive Power BI dashboard.

Although machine learning models provide accurate predictions, decision-makers often require visual representations of data to understand trends and make informed business decisions. Therefore, Microsoft Power BI was used to create an interactive dashboard that transforms complex analytical results into meaningful business insights.

The dashboard was designed with a user-friendly interface consisting of four pages. Each page focuses on a different aspect of customer churn analysis, allowing users to explore customer behavior through dynamic visualizations, KPI cards, charts, slicers, and filters.

13.2 Objectives of the Dashboard

The main objective of the Power BI dashboard was to provide business users with an interactive platform for monitoring customer churn and understanding the factors influencing customer retention.

The dashboard was designed to:

- Present key business metrics through KPI cards.
- Analyze customer demographics and churn behavior.
- Visualize customer service usage patterns.
- Monitor revenue and customer lifetime value.
- Support business decision-making through interactive reports.
- Enable users to filter data using slicers and drill-down features.

The dashboard transforms raw analytical outputs into business-friendly visualizations that can be easily interpreted by managers and stakeholders.

13.3 Dashboard Design

The dashboard follows a clean and consistent design throughout all pages. A common color theme, standardized charts, and uniform formatting were applied to improve readability and maintain a professional appearance.

Interactive slicers were added to allow users to filter the dashboard by different customer categories without modifying the underlying dataset. KPI cards were placed at the top of the pages to provide quick summaries of important business metrics.

The dashboard was developed with the goal of balancing visual appeal and analytical depth while ensuring that users can quickly identify important trends and patterns.

13.4 Dashboard Pages

The Power BI dashboard consists of four interactive pages, each focusing on a specific area of customer churn analysis.

Page 1 – Executive Overview

The Executive Overview page provides a high-level summary of the entire customer dataset. It serves as the starting point for business users by displaying key performance indicators and overall customer distribution.

The page includes KPI cards that summarize:

- Total Customers
- Churned Customers
- Retained Customers
- Churn Rate
- Average Monthly Charges
- Average Customer Tenure

In addition to the KPI cards, the page includes visualizations that present customer distribution based on gender, contract type, and churn status.

Interactive slicers enable users to filter the dashboard by customer attributes and instantly update all visualizations.

This page provides decision-makers with a quick overview of business performance and customer retention.

Page 2 – Customer Demographics and Contract Analysis

The second page focuses on understanding how customer demographics and contract-related factors influence churn.

The visualizations on this page include:

- Churn by Gender
- Churn by Senior Citizen
- Churn by Partner Status
- Churn by Dependents
- Churn by Internet Service
- Churn by Contract Type
- Churn by Payment Method

These visualizations help identify customer groups that are more likely to discontinue their services.

The analysis indicates that customers with month-to-month contracts and certain payment methods experience higher churn compared to customers with long-term contracts.

This page enables businesses to identify high-risk customer groups and design targeted retention strategies.

Page 3 – Service Usage and Customer Behaviour Analysis

The third page examines how customer service usage affects customer retention.

The dashboard includes visualizations based on:

- Online Security
- Online Backup
- Device Protection
- Tech Support
- Streaming TV
- Streaming Movies
- Customers by Tenure Group
- Churn Rate by Tenure Group
- Monthly Charges vs Tenure Scatter Plot

These visualizations reveal how different services influence customer behavior.

Customers who subscribe to value-added services such as Online Security and Tech Support generally demonstrate stronger customer loyalty and lower churn rates.

The scatter plot comparing Monthly Charges and Tenure provides additional insight into the relationship between customer spending and retention.

This page helps businesses identify opportunities to improve customer satisfaction through service offerings.

Page 4 – Revenue and Customer Lifetime Value Analysis

The final dashboard page focuses on the financial impact of customer churn.

The page includes the following visualizations:

- Revenue by Churn Status
- Monthly Charges Distribution
- Total Charges Distribution
- Revenue by Payment Method
- Monthly Charges by Contract Type
- Monthly Charge Band Analysis
- Churn Rate by Monthly Charge Band
- Customer Lifetime Value Analysis
- Customer Segmentation by Bronze, Silver, Gold, and Platinum Categories

These visualizations help businesses understand how churn affects revenue and identify high-value customer segments.

The Customer Lifetime Value analysis enables organizations to prioritize customers who generate greater long-term business value.

13.5 KPI Cards

Several KPI cards were developed to provide quick access to important business metrics.

The dashboard includes KPIs such as:

- Total Customers

- Total Churned Customers
- Total Retained Customers
- Customer Churn Rate
- Average Monthly Charges
- Average Customer Tenure
- Total Revenue
- Average Customer Lifetime Value

These KPI cards provide a concise summary of business performance and support quick decision-making.

13.6 DAX Measures

Data Analysis Expressions (DAX) were used extensively throughout the dashboard to create calculated measures and support dynamic visualizations.

Several DAX measures were developed, including:

- Total Customers
- Churned Customers
- Retained Customers
- Churn Rate
- Total Revenue
- Average Monthly Charges
- Average Tenure
- Customer Lifetime Value
- Revenue by Customer Segment

These measures ensure that all visualizations update automatically when users interact with filters or slicers.

13.7 Interactive Features

The dashboard includes multiple interactive features that improve usability and provide a better analytical experience.

These features include:

- Interactive slicers
- Dynamic filtering

- Drill-down functionality
- Cross-filtering between visuals
- Dynamic KPI updates
- Responsive charts
- Consistent navigation across pages

These features allow users to explore customer data from different perspectives without requiring additional reports.

13.8 Business Insights

The Power BI dashboard generated several important business insights.

Some of the key findings include:

- Customers with month-to-month contracts exhibit the highest churn rates.
- Long-term contract customers demonstrate better customer retention.
- Customers with longer tenure are significantly less likely to churn.
- Higher monthly charges are associated with increased customer churn.
- Customers without Online Security or Tech Support services are more likely to discontinue their subscriptions.
- Electronic Check users experience relatively higher churn compared to customers using other payment methods.
- Fiber Optic Internet customers show higher churn rates than DSL customers.
- Customers with higher Lifetime Value should be prioritized for retention initiatives.
- Platinum and Gold customers contribute a significant portion of overall revenue and should receive personalized retention strategies.

These insights help organizations develop proactive customer retention plans and optimize business performance.

13.9 Benefits of the Dashboard

The completed Power BI dashboard offers several benefits for business users.

Some of the major advantages include:

- Easy visualization of customer churn trends.
- Interactive analysis of customer behavior.
- Revenue monitoring through dynamic reports.

- Customer Lifetime Value analysis.
- Identification of high-risk customers.
- Better customer segmentation.
- Improved business decision-making.
- Professional business intelligence reporting.

The dashboard enables managers to quickly identify areas requiring attention and supports strategic planning using real-time analytical insights.

13.10 Outcome of Week 4

The final phase of the project successfully transformed analytical and machine learning results into a comprehensive business intelligence solution.

An interactive four-page Power BI dashboard was developed to visualize customer churn, customer demographics, service usage, revenue, and Customer Lifetime Value. Dynamic KPI cards, DAX measures, slicers, and interactive charts were implemented to enhance usability and improve data exploration.

By the end of Week 4, the following objectives were successfully achieved:

- Developed a professional four-page Power BI dashboard.
- Created dynamic KPI cards.
- Implemented DAX measures for business metrics.
- Designed interactive visualizations for customer analysis.
- Presented Customer Lifetime Value and customer segmentation.
- Added slicers, filters, and drill-down functionality.
- Generated actionable business insights.
- Completed the final business intelligence layer of the project.

The completion of this phase marked the successful conclusion of the Customer Churn Prediction & Lifetime Value (LTV) Engine. The project now provides an end-to-end solution that combines data preprocessing, machine learning, API deployment, and business intelligence into a single integrated system capable of supporting data-driven customer retention strategies.

CHAPTER 14

CHALLENGES FACED

Throughout the development of this project, I encountered several technical and analytical challenges. Each challenge provided an opportunity to improve my understanding of data analytics, machine learning, API development, and business intelligence.

One of the initial challenges was preparing the dataset for analysis. The raw dataset contained missing values, inconsistent data types, and duplicate records that needed to be cleaned before any meaningful analysis could be performed. Ensuring data quality required careful preprocessing and validation.

Another challenge was selecting the most relevant features for machine learning. The dataset contained several categorical variables that required encoding before they could be used by the algorithms. Choosing appropriate preprocessing techniques while preserving the information contained in the dataset was an important step.

Model selection was another significant challenge. Different machine learning algorithms produced different prediction results, making it necessary to compare multiple models using various evaluation metrics rather than relying solely on accuracy. Understanding the strengths and limitations of each model helped in selecting the most suitable algorithm for deployment.

Deploying the model using FastAPI introduced new concepts related to REST API development. Learning how to structure API endpoints, validate input data, and test the application using Swagger UI required additional effort beyond traditional machine learning workflows.

The final challenge involved designing an interactive Power BI dashboard. Initially, some visualizations became overcrowded due to the large number of customer records. Selecting appropriate chart types, improving dashboard layout, creating DAX measures, and maintaining consistency across all pages required several iterations before achieving the final design.

Despite these challenges, each stage of the project enhanced my practical skills and provided valuable experience in solving real-world business problems.

CHAPTER 15

SKILLS APPLIED

This project allowed me to apply knowledge from multiple areas of data science, machine learning, software development, and business intelligence.

Programming Languages

- Python
- DAX

Data Analysis

- Data Cleaning
- Data Preprocessing
- Exploratory Data Analysis (EDA)
- Statistical Analysis
- Feature Engineering

Machine Learning

- Logistic Regression
- Decision Tree
- Random Forest
- Model Training
- Model Evaluation
- Classification
- Customer Churn Prediction

Business Analytics

- Customer Lifetime Value (LTV)
- Customer Segmentation
- Business Intelligence
- Customer Behaviour Analysis

API Development

- FastAPI
- REST API
- JSON
- Swagger UI
- Pickle

Data Visualization

- Power BI
- Interactive Dashboards
- KPI Cards
- Charts and Graphs
- Dashboard Design

Version Control

- Git
- GitHub

CHAPTER 16

LEARNING OUTCOMES

This project provided me with practical experience in developing an end-to-end data analytics solution. It strengthened both my technical knowledge and problem-solving skills by allowing me to work through every stage of a real-world machine learning project.

Through this project, I learned how to:

- Understand business problems using data analytics.
- Clean and preprocess real-world datasets.
- Perform Exploratory Data Analysis (EDA).
- Apply feature engineering techniques.
- Build and evaluate machine learning classification models.
- Compare multiple machine learning algorithms.
- Estimate Customer Lifetime Value (LTV).
- Perform customer segmentation based on business value.
- Deploy machine learning models using FastAPI.
- Develop REST APIs for real-time predictions.
- Create professional Power BI dashboards.
- Develop DAX measures for dynamic reporting.
- Present analytical findings through interactive visualizations.
- Manage project versions using Git and GitHub.
- Document technical work in a structured and professional manner.

This project also improved my ability to think analytically, troubleshoot technical issues, and translate complex data into meaningful business insights.

CHAPTER 17

FUTURE SCOPE

Although the project successfully achieved its objectives, several enhancements can be implemented in the future to improve its capabilities.

Possible future improvements include:

- Integrating real-time customer data from databases.
- Deploying the application on cloud platforms such as Microsoft Azure or AWS.
- Developing a web application for customer churn prediction.
- Building automated retraining pipelines for machine learning models.
- Incorporating deep learning techniques for improved prediction accuracy.
- Performing sentiment analysis using customer feedback and reviews.
- Integrating email or SMS alerts for high-risk customers.
- Adding real-time streaming dashboards using Power BI Service.
- Expanding the solution to other industries such as banking, healthcare, insurance, and e-commerce.
- Implementing recommendation systems that suggest personalized customer retention strategies.

These enhancements would make the solution more scalable and suitable for enterprise-level business environments.

CHAPTER 18

CONCLUSION

The **Customer Churn Prediction & Lifetime Value (LTV) Engine** project successfully demonstrates the complete lifecycle of a modern data analytics and machine learning solution. The project began with collecting and preparing customer data through preprocessing and exploratory data analysis, followed by feature engineering and the development of predictive machine learning models. Customer Lifetime Value was estimated to understand the long-term contribution of each customer, and customers were segmented into different value groups to support better business decisions.

The trained machine learning model was then deployed using FastAPI, allowing customer churn predictions to be accessed through REST API endpoints. Finally, an interactive Power BI dashboard was developed to present customer behavior, churn trends, revenue analysis, and customer lifetime value in a visually meaningful way.

This project highlights how machine learning, API development, and business intelligence can work together to solve real-world business problems. The developed solution enables organizations to identify customers who are likely to churn, estimate their future value, and take proactive steps to improve customer retention.

Working on this project has significantly improved my understanding of data preprocessing, machine learning, model deployment, dashboard development, and business analytics. It has also strengthened my ability to work on complete end-to-end projects that integrate multiple technologies into a single solution.

Overall, this internship project has been an excellent learning experience and has enhanced both my technical expertise and practical understanding of solving business problems through data-driven decision-making.

REFERENCES

The following resources were referred to during the development of this project:

- Python Software Foundation. *Python Documentation*. https://docs.python.org/
- Pandas Documentation. https://pandas.pydata.org/docs/
- NumPy Documentation. https://numpy.org/doc/
- Matplotlib Documentation. https://matplotlib.org/stable/
- Scikit-learn Documentation. https://scikit-learn.org/stable/
- FastAPI Documentation. https://fastapi.tiangolo.com/
- Microsoft Power BI Documentation. https://learn.microsoft.com/power-bi/
- Microsoft DAX Documentation. https://learn.microsoft.com/dax/
- Git Documentation. https://git-scm.com/doc
- GitHub Documentation. https://docs.github.com/

APPENDIX

The following project files are included in the GitHub repository:

- Data Cleaning and Preprocessing Notebook
- Exploratory Data Analysis Notebook
- Machine Learning Model Notebook
- Customer Lifetime Value (LTV) Analysis
- Customer Segmentation
- FastAPI Application
- Trained Machine Learning Model
- Power BI Dashboard (.pbix)
- Project Documentation
- GitHub Repository

FINAL PROJECT SUMMARY

Project Title: Customer Churn Prediction & Lifetime Value (LTV) Engine

Project Duration: 4 Weeks

Week 1

- Data Collection
- Data Cleaning
- Data Preprocessing
- Exploratory Data Analysis

Week 2

- Feature Engineering
- Machine Learning
- Customer Lifetime Value (LTV)
- Customer Segmentation

Week 3

- Model Deployment using FastAPI
- REST API Development
- Swagger UI Testing

Week 4

- Interactive Power BI Dashboard
- DAX Measures
- Business Intelligence
- Data Visualization

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- FastAPI
- Pickle
- Power BI
- DAX
- Git
- GitHub

Project Deliverables

- Cleaned Dataset
- Exploratory Data Analysis
- Machine Learning Models
- Customer Churn Prediction Engine
- Customer Lifetime Value (LTV) Engine
- Customer Segmentation
- FastAPI REST API
- Interactive Power BI Dashboard
- Complete Project Documentation
- GitHub Repository
